Fall 2025-26

# BMAT205L - Discrete Mathematic And Graph Theory

## Tutorial - 7 (module 5, 6)

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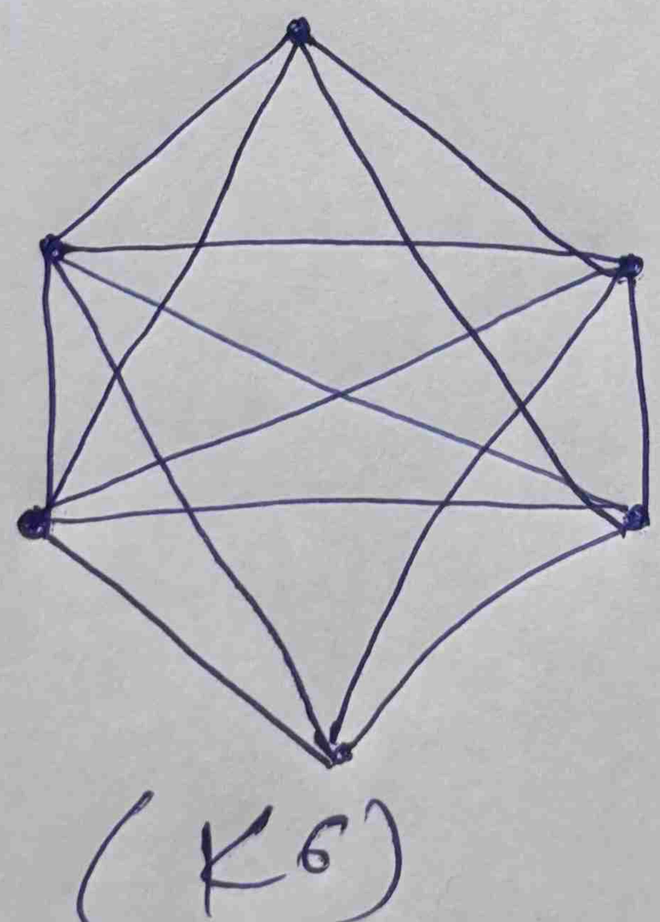
Qve-1 Draw the Graphs K6 and K5,3

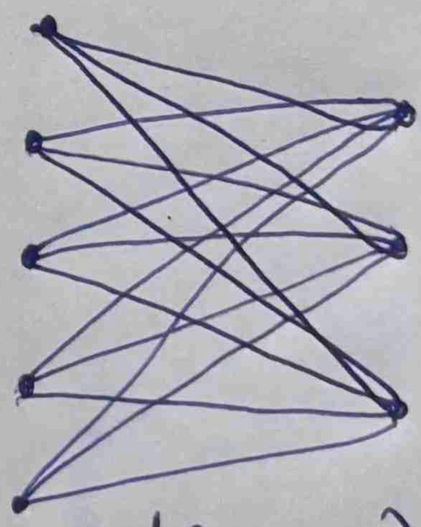
Solve :- K6 :-

- Vertices = 6
- Edges = $\frac{6(6-1)}{2}$ = 15
- Each vertex is connected to every other vertex.

(K, 5,3)



(K6)



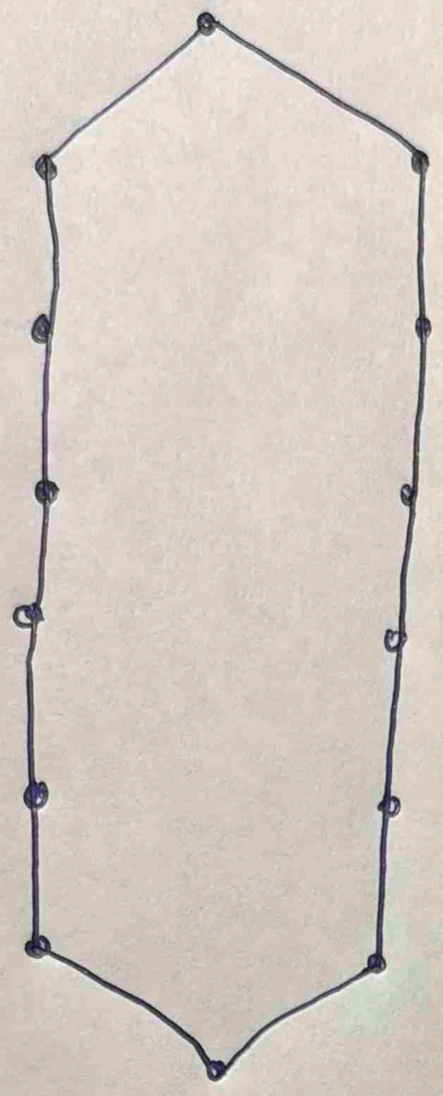
(K5,3)

Vertices = 8

Edges = 5×3 = 15

Que-2 Give An Example of a graph with 11 vertices such that (2)

(i) Eulerian And Hamiltonian.



Both Hamiltonian And Eulerian

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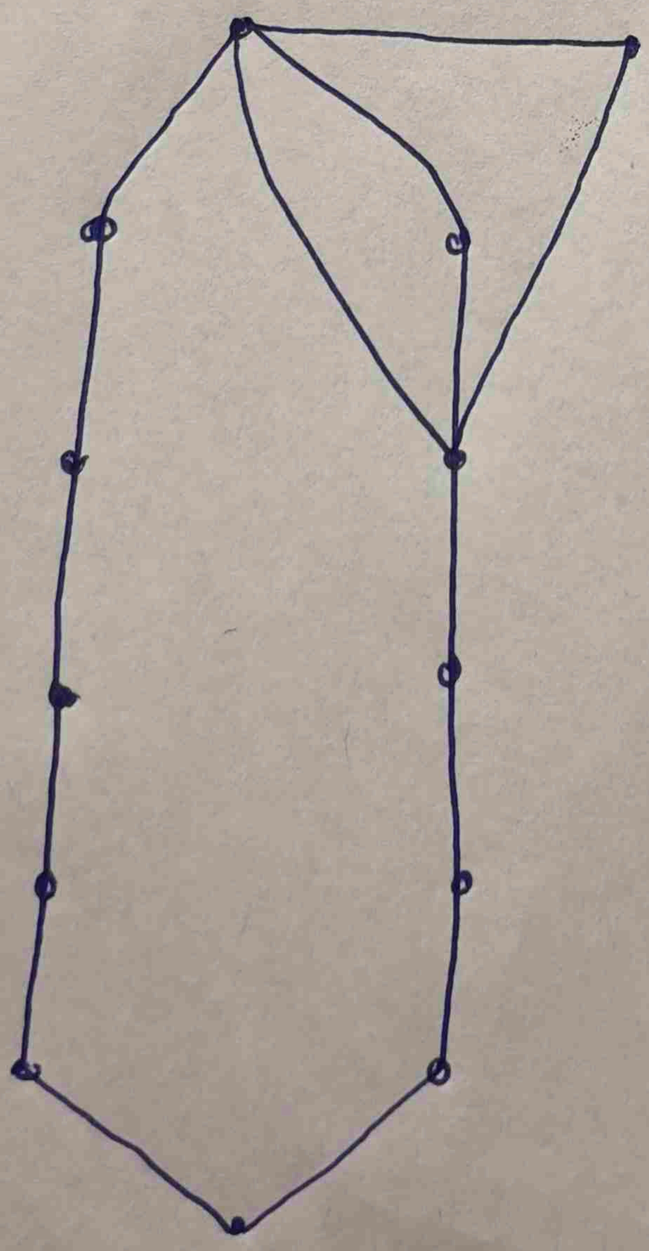
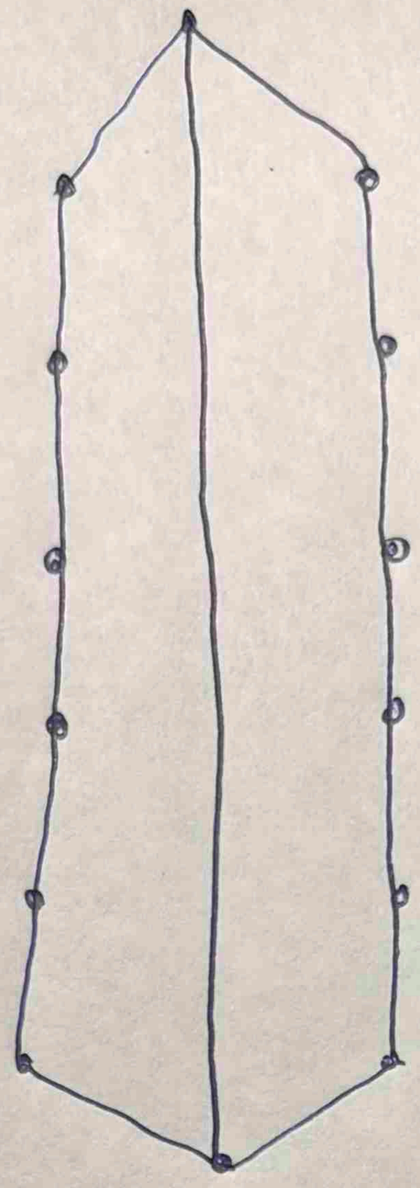
(ii) Eulerian But not Hamiltonian



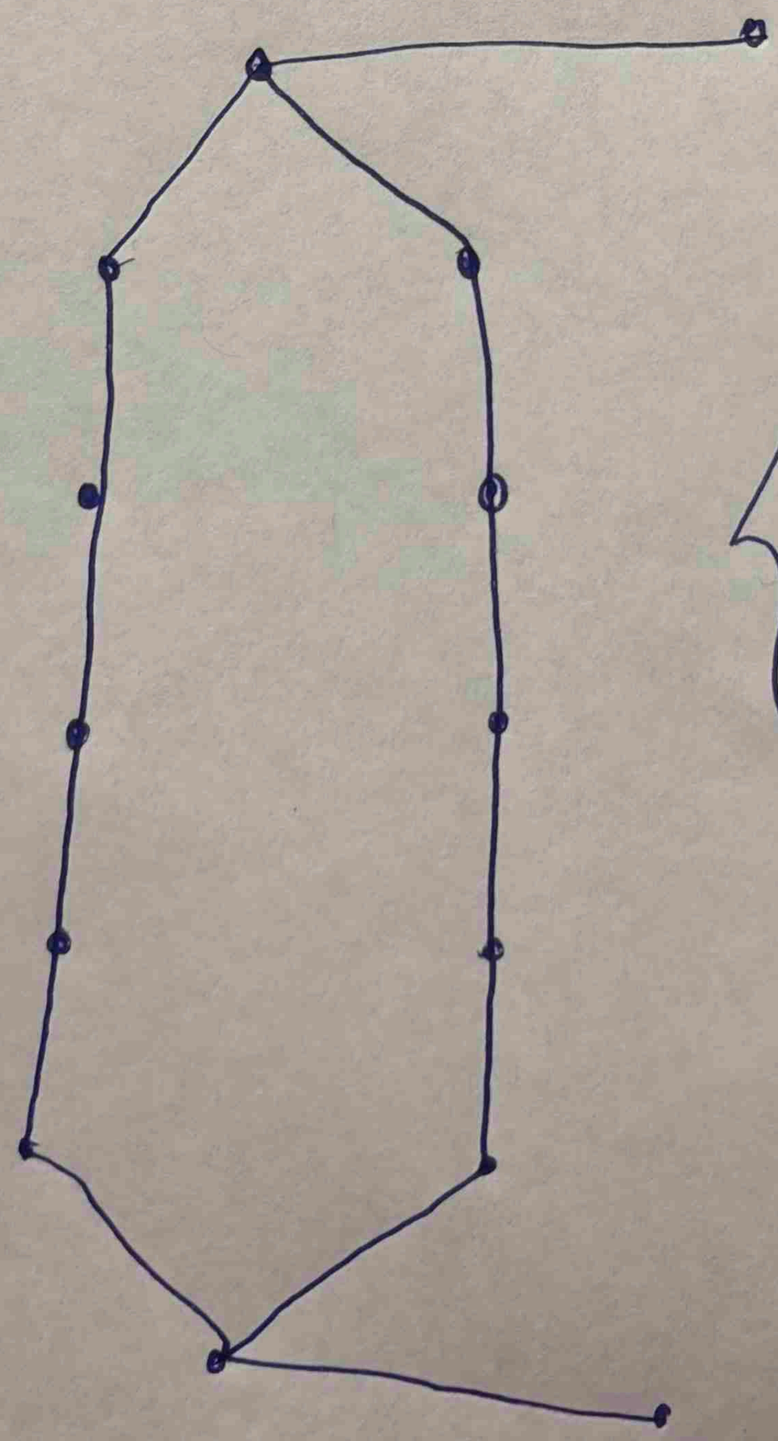
Eulerian But not Hamiltonian

(iii) Not Eulerian But Hamiltonian :-



{Hamiltonian But not Eulerian}

(iv) Not Eulerian Not Hamiltonian



{Not Eulerian And not a Hamiltonian}

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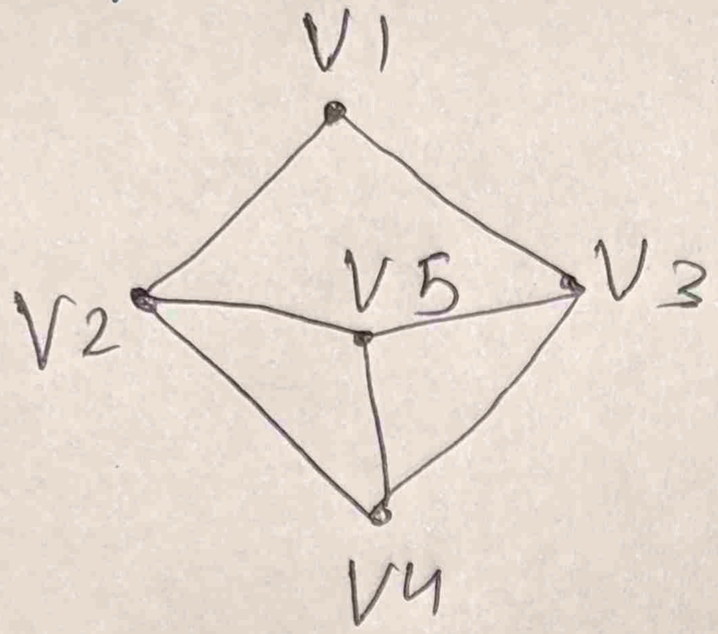
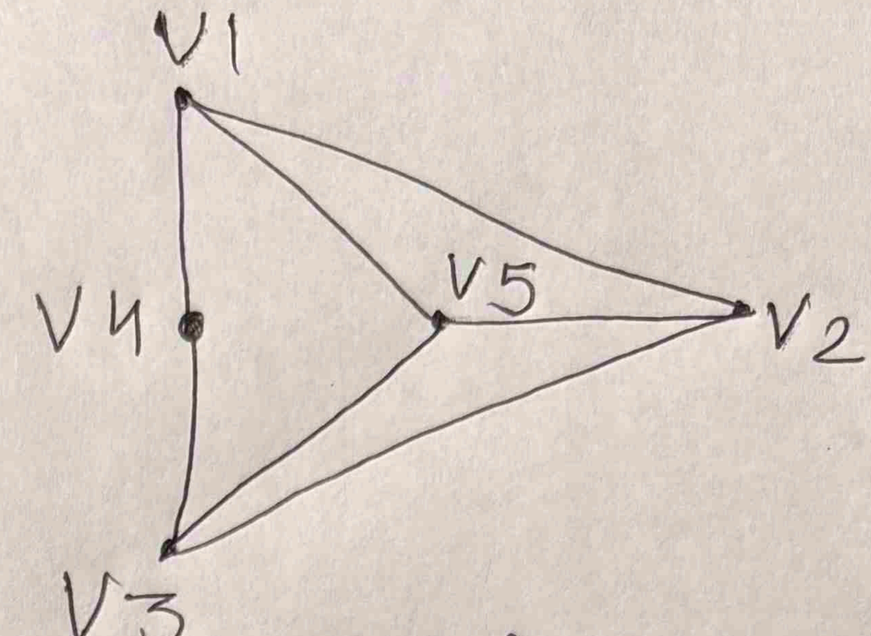
Que-3 Check whether the following two graph Isomorphic or not (4)

(i)




(G1)

(ii)




(G2)

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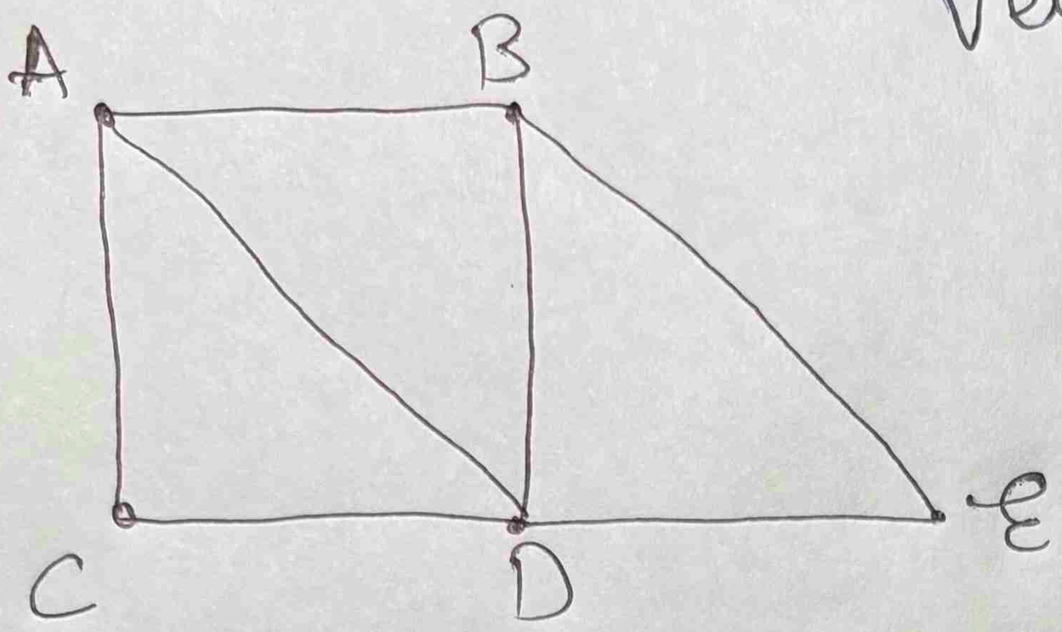
⇒ Two Graphs are G1 and G2 F
- They have the same number of And Edges.
- The list of vertices degree is
- There exists a one to one cor b/w vertices preserving Adjacency.

⇒ Suppose Both graphs have Degree Seawence (3,3,2,2,2,2)

⇒ If Adjacency matches under relabeling Hence, they are Isomorphic (V)

Que-4 find the Euler path or an Euler circuit, if it exists in each of the following three Graphs. If it does not exist. Explain why? (5)

(i)




(G1)

→ 4 odd - Degree vertices

Vertices forming a Square (A, B, C, D)

{Diagonals AC and BD}

Deg (A) =
Deg (B) =
Deg (C) =
Deg (D) =

Conclusion :- No Euler path or C
too many odd de

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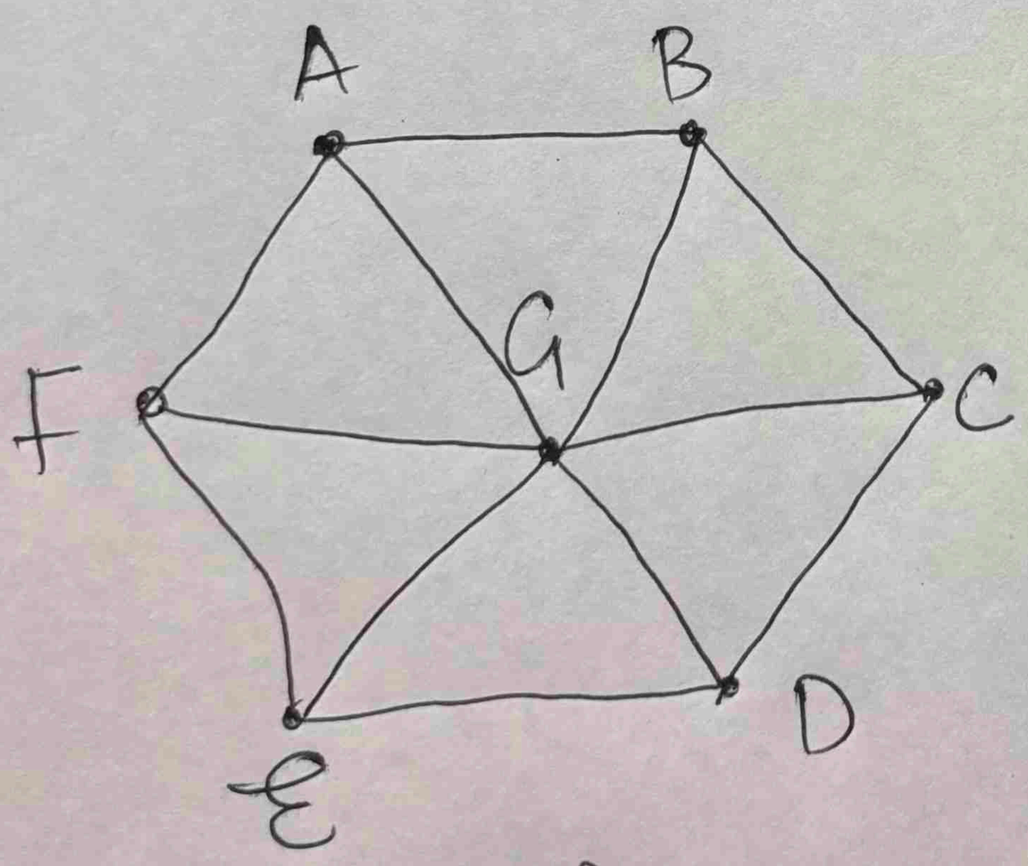
(ii)




(G2)

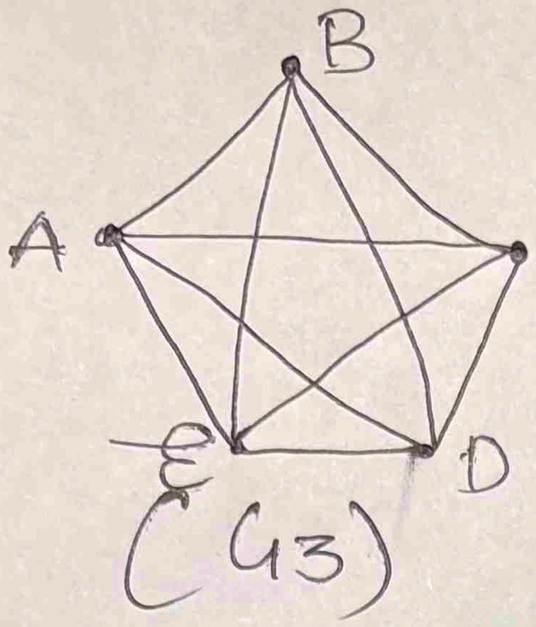
Degrees :-

deg (A) = 3
deg (B) = 2
deg (C) = 3
deg (D) = 2

• Odd Vertices = A, C (2 odd)

• Conclusion :- Euler path Exists, But not a Euler circuit.

(iii)

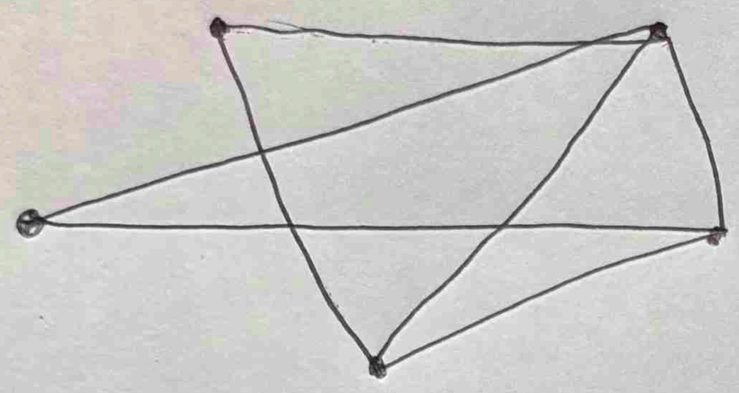



(G3)

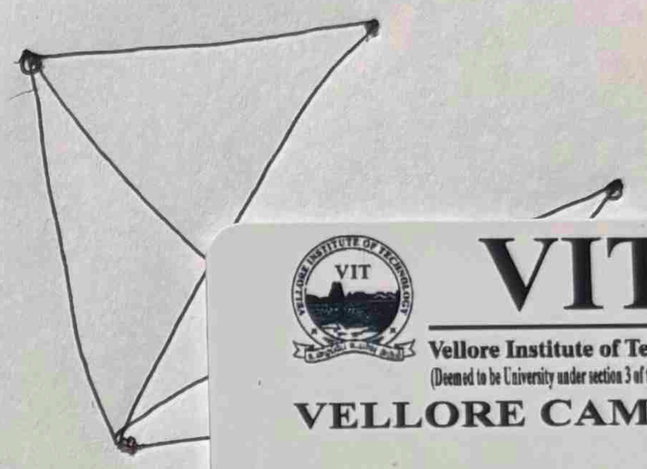
- Hexagon (6-cycle) ⑥
- Every Vertex degree = 2 (even)
- Conclusion :- Euler Circuit exists (and Euler path too)

Que-5 Check whether the following Graphs are Isomorphic or not.

(i)



(ii)



Solve :- Q1 :- Square with one-Di

Vertices A-B-C-D with o

→ Edges = AB, CD, BC, DA, A

Q2 :- Another four vertex graph

PQ, RS, QR, SP, PR

① Degree Sequence :- (3,3,2,2)
Identical

② number of vertices :- 4, edges = 5 in both

All Edges Preserved ⇒ { Isomorphic Graphs } Ans

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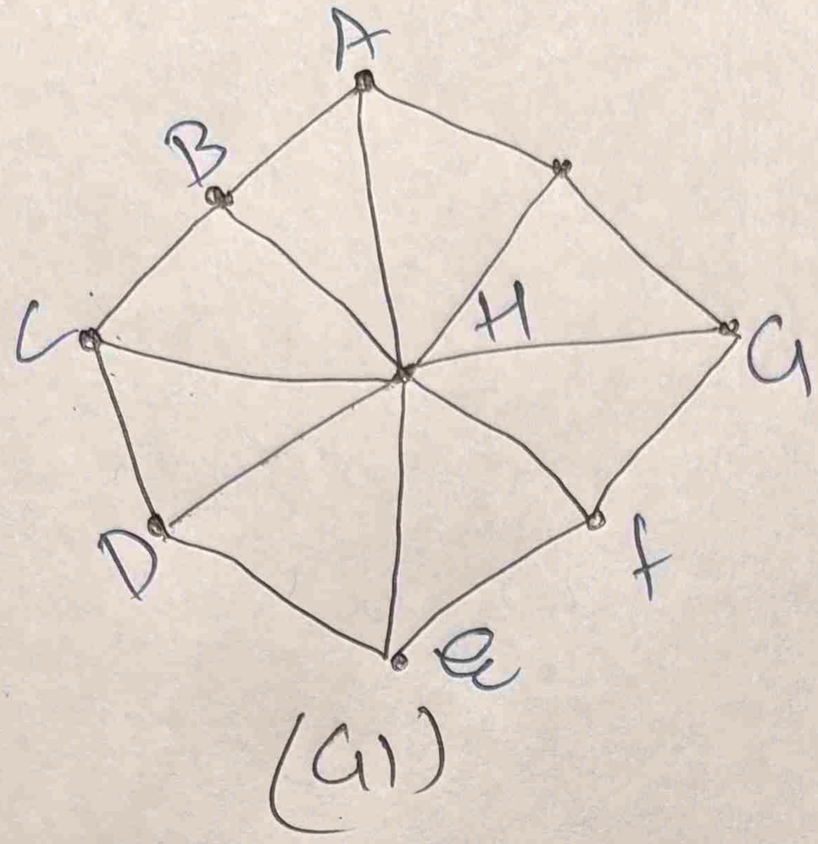
Que-7 Check whether the following two graphs are Isomorphic or not

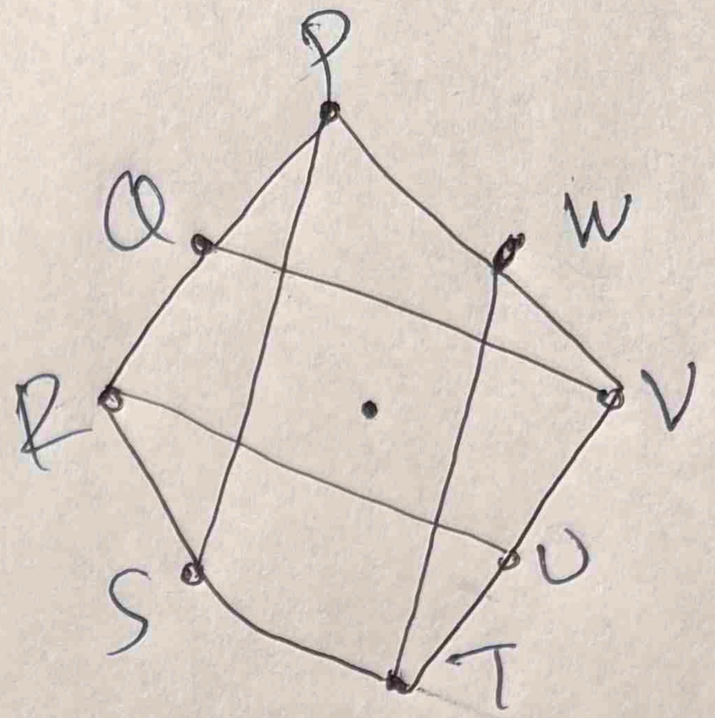
(4)

(i)




(G1)

(ii)




(G2)

Sowe : (G1) :- Vertices = A, B, C, D, E, F, G, H

Edges = AB, BC, CD, DA

Edges = 6

Degree of each vertex = 3

So Degree Seawencus = (3, 3, 3,

(G2) :- Vertices :- P, Q, R, S, T, U,

Edges :- PQ, QR, RS, ST,

VW, QV, RU, SP, TW

Edges :- 11

Even Before checking Adjacency, they differ in degree seawencus. Cannot be Isomorphic

Conclusion :- { The two graphs are not Isomorphic Any }

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Que-9 A connected planar Graph contains 10 vertices and vertics divides the planes into seven Regions. Compute the number of Edges in the Graph. Give An Example of such a Graph. (8)

Solve :- for Any connected planar Graph,

$$\{V - E + F = 2\}$$

V = no of vertics
E = no of Edges
F = no of faces (Regions)

Given :- V = 10, F = 7

Substitute into Euler form

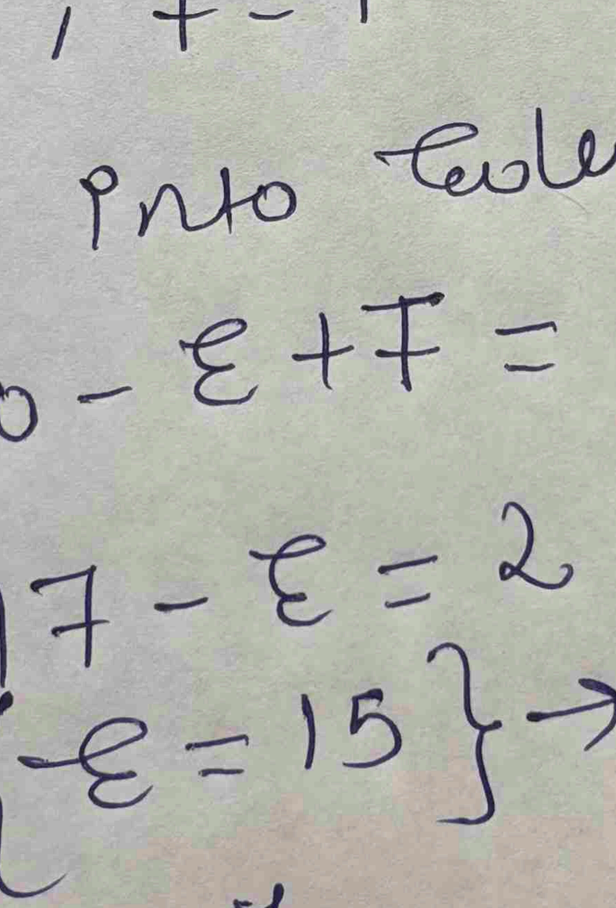
$$10 - E + 7 = 2$$

Simplify :-

$$17 - E = 2$$

$$\{E = 15\} \rightarrow \text{No of Edges}$$

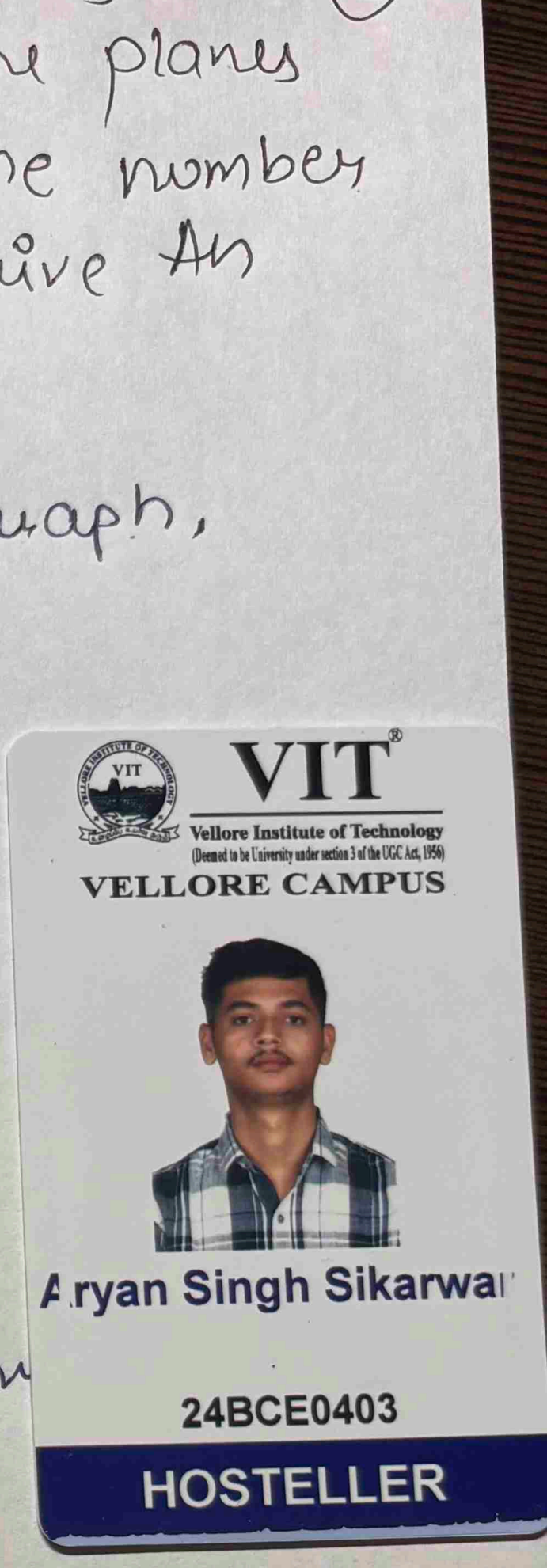
Graph :-



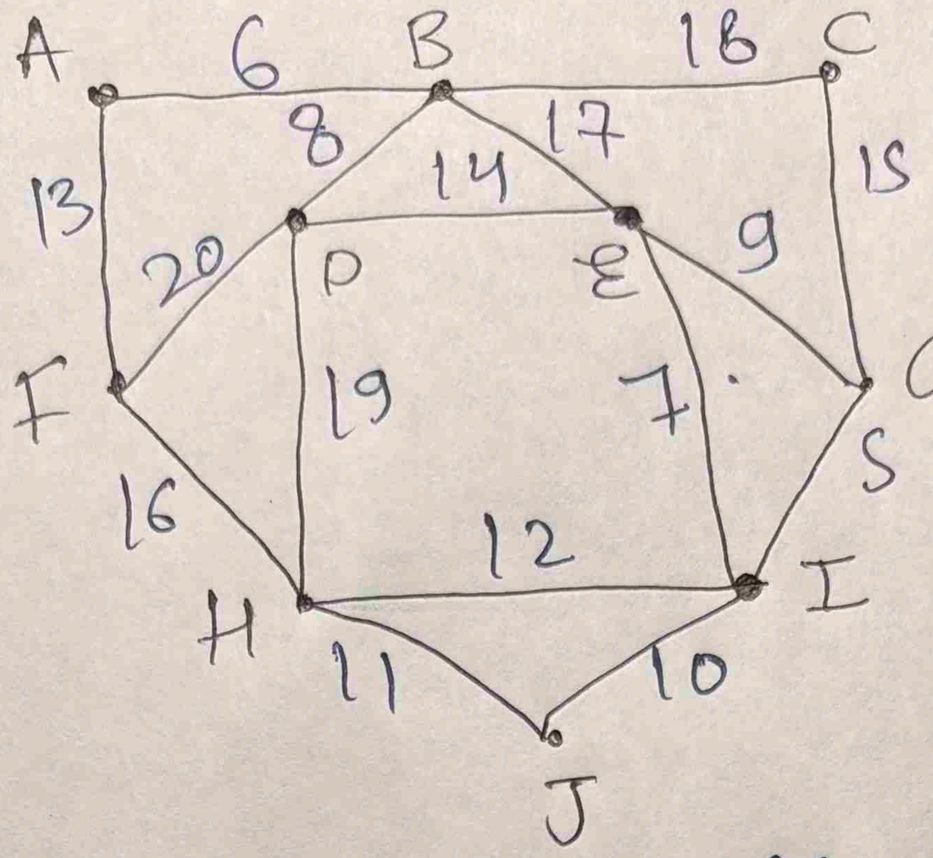

Graph with :-

$$\left\{\begin{array}{l} V = 10 \\ E = 15 \\ F = 7 \end{array}\right\}$$

Ans



Qve-10 Using Kruskals Algorithm, find the minimum Spanning Tree for the following weighted, undirected Graph. (9)




Vertices :-

{ A, B, C, D, E, F, G, H, I, J }

Sowe :- Sort Edgs in Ascending weight :-

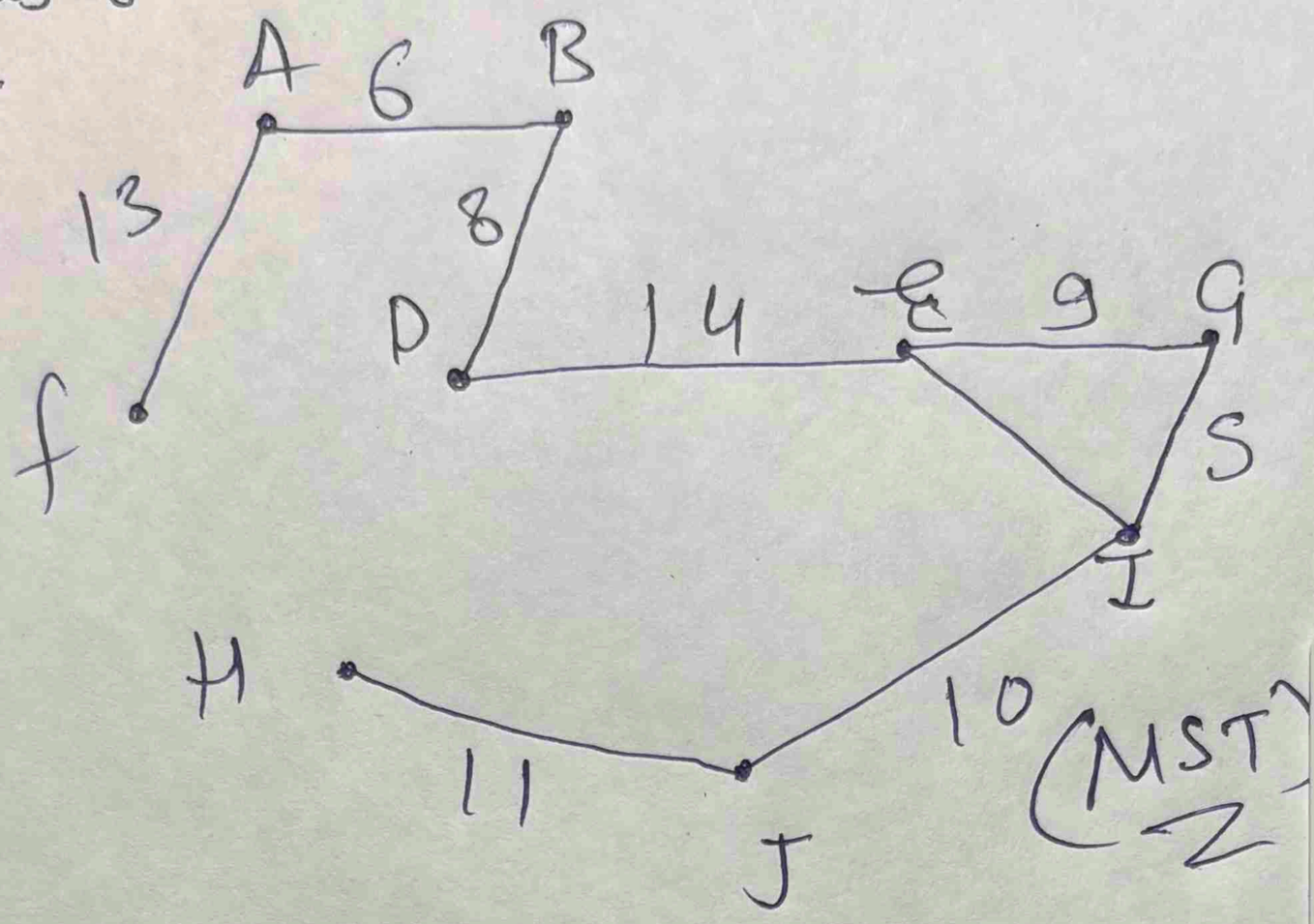
| ORDER | EDGE | WE |
|---|---|---|
| 1 | G-I | 5 |
| 2 | A-B | 6 |
| 3 | E-I | 7 |
| 4 | B-D | 8 |
| 5 | E-G | 9 |
| 6 | I-J | 10 |
| 7 | H-J | 11 |
| 8 | H-I | 12 |
| 9 | F-A | 13 |
| 10 | D-E | 14 |
| 11 | C-G | 15 |
| 12 | F-H | 16 |
| 13 | B-E | 17 |
| 14 | C-B | 18 |
| 15 | D-H | 19 |
| 16 | F-D | 20 |

Now, we can have a spanning tree

$V - 1 = Edges$

$\{10 - 1 = 9\ Edges\}$

MST Edges :-




(MST)

Total weights :-

$5+6+7+8+9+10+11+12+13+$

Final Verdict :-

Vertices = 10

Edges = 9

Total weight = 83

Edges included = G-I, A-B, E-I, B-D, E-G, I-J, H-J, F-A, D-E

Ans

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Que-11 for what value of n, the compute Graph Kn be phanar? Justify why?

Solve :- A planar Graph can be drawn on a plane without Any Edge Crossing.

• A complete Graph Kn has :-

{ number of Edges = $\frac{n(n-1)}{2}$ }

for connected Graph :- { $V - E + F = 2$ }

Also, the simple Graphs with

{ $E \leq 3n-6$ }

(This follows for every face at least 3 edges.

Apply the planar Graph Condition

for Kn :- $\frac{n(n-1)}{2} =< 3n-6$

multiply both side by 2 :- $n(n-1) \leq 6n-12$

$n^2 - n \leq 6n-12$

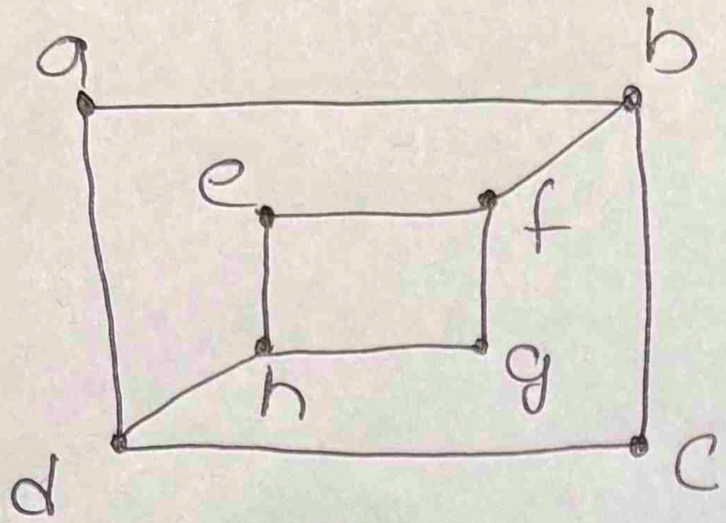
$n^2 - 7n + 12 \leq 0$

$(n-3)(n-4) \leq 0$

{ so, the Inequality holds } $\Rightarrow$ { $3 \leq n \leq 4$ } Kn is planar

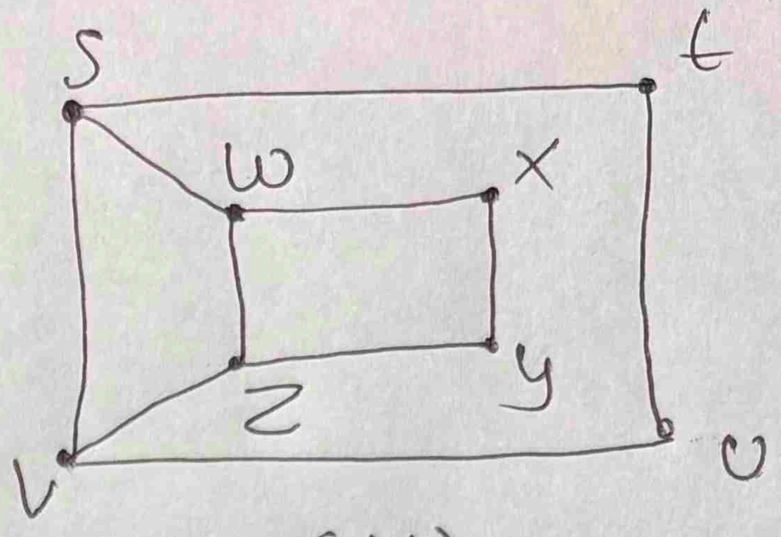
Que-13 Check whether the following two graphs are isomorphic or not. (12)

(i)




(G)

(ii)




(H)

Sol^n :- Graph (G) has vertices :-

a, b, c, d, e, f, g, h

Graph (H) has vertices :-

s, t, u, v, w, x, y, z

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- Graph G :-

① outer Rectangle :- a-b-c-d-a
② inner Rectangle :- e-f-g-h-e
③ extra Edges :- b-g and d-h

- Graph H :-

① outer Rectangle :- s-t-u-v-s
② inner Rectangle :- w-x-y-z-w
③ extra Edges :- b-g and d-h

⇒ So both have :-

① 8 vertices
② 10 Edges (4 outer + 4 inner + 2 extra)

Step-2 Since all the identical degree Seawences and Counts are present in both.

Step-3 All adjacency mateh only this mapping :-

outer :- a-b-c-d-a ⟷ s-t-u-v-s

inner :- e-f-g-h-e ⟷ w-x-y-z-w

cross links :- b-g ⟷ t-x, d-h ⟷ v-z

{Adjacency preserved}

Final conclusion ⇒ {Graph (G) and (H) Isomorphic}

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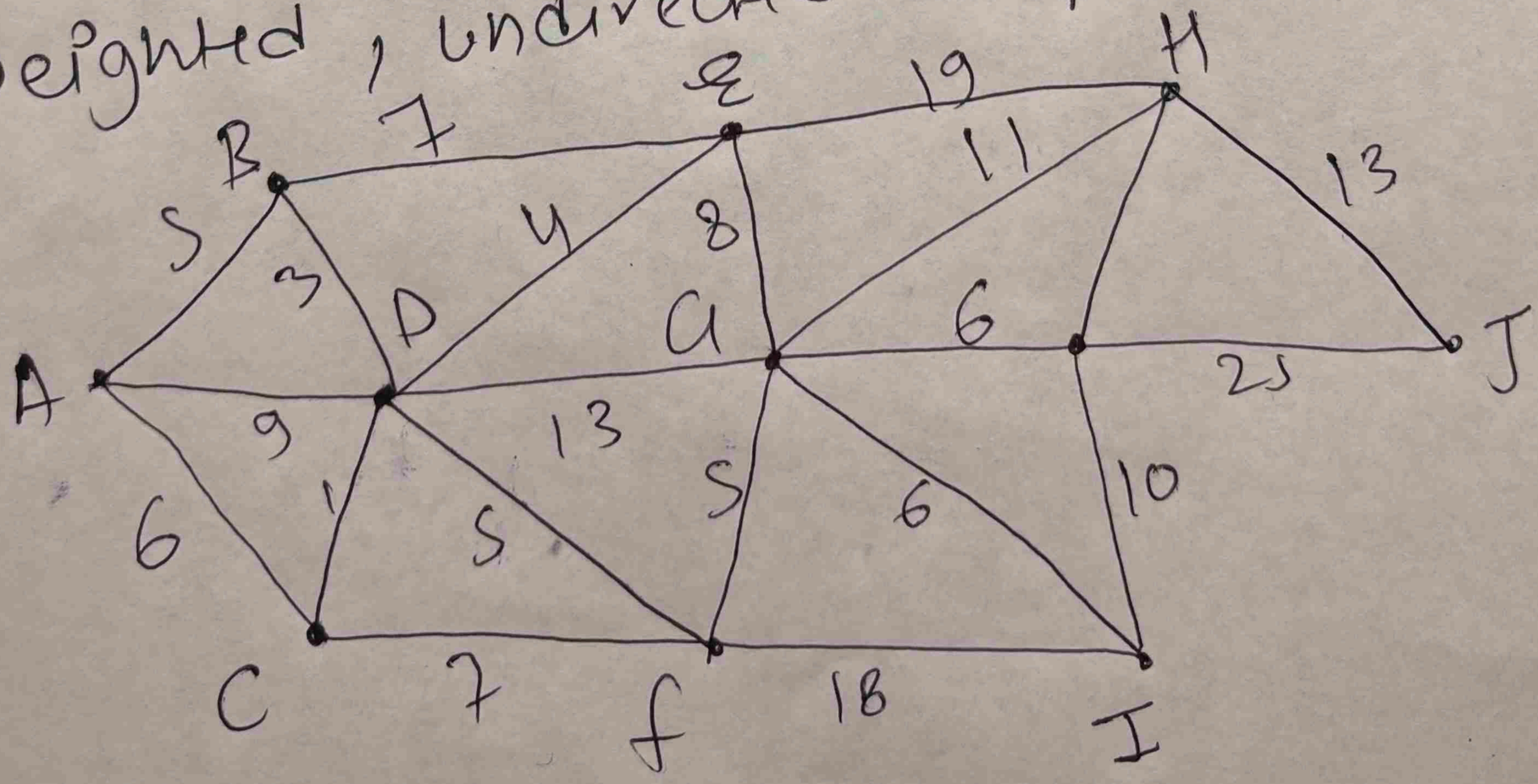
Qve-15 Using Kruskal's Algorithm, find the minimum Spanning tree for the follow weighted, undirected Graph.



Sowe :- Vertics :- {A, B, C, D, E, F, G, H, I, J}

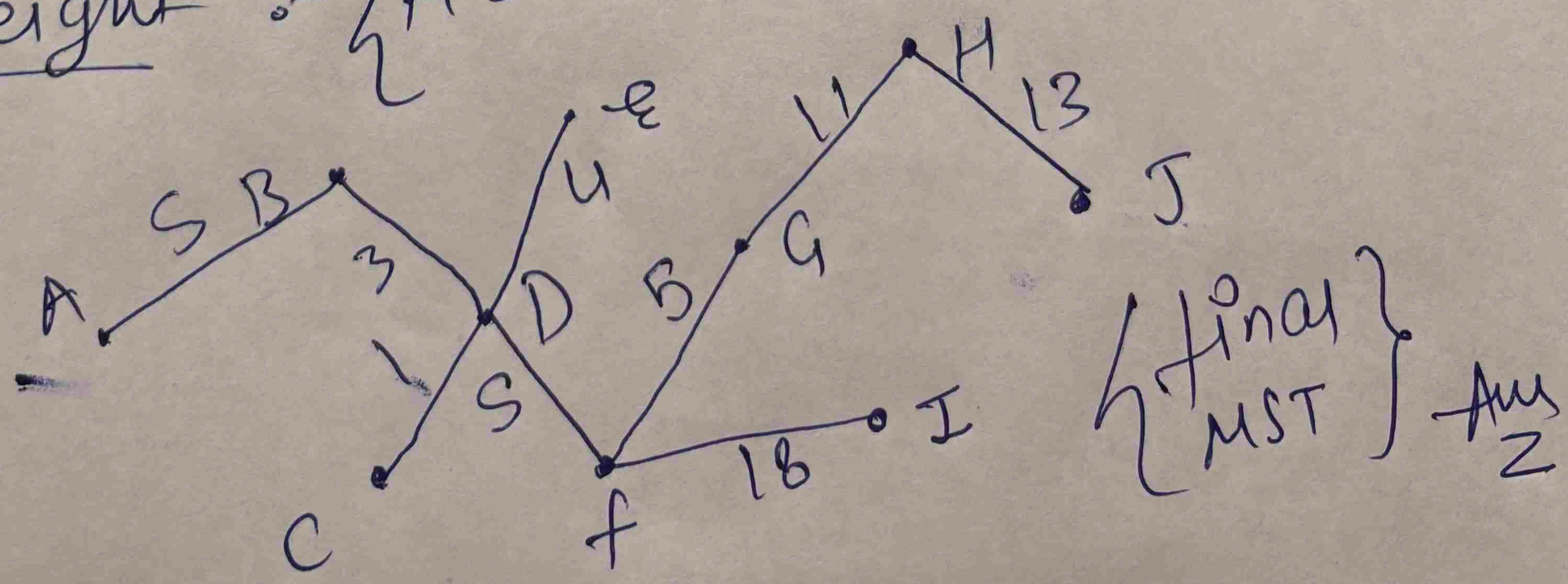
Step-1 :- Sort Edges By weight (Ascending) :-

| ORDER | EDGES | WEIGHT |
|---|---|---|
| 1 | C-D | 1 |
| 2 | B-D | 3 |
| 3 | D-E | 4 |
| 4 | A-B | 5 |
| 5 | D-F | 5 |
| 6 | A-C | 6 |
| 7 | F-G | 7 |
| 8 | C-F | 8 |
| 9 | E-G | 9 |
| 10 | A-D | 11 |
| 11 | G-H | 11 |
| 12 | H-J | 13 |
| 13 | D-G | 13 |
| 14 | F-I | 18 |
| 15 | E-H | 19 |
| 16 | G-J | 25 |

Final MST Edges :-

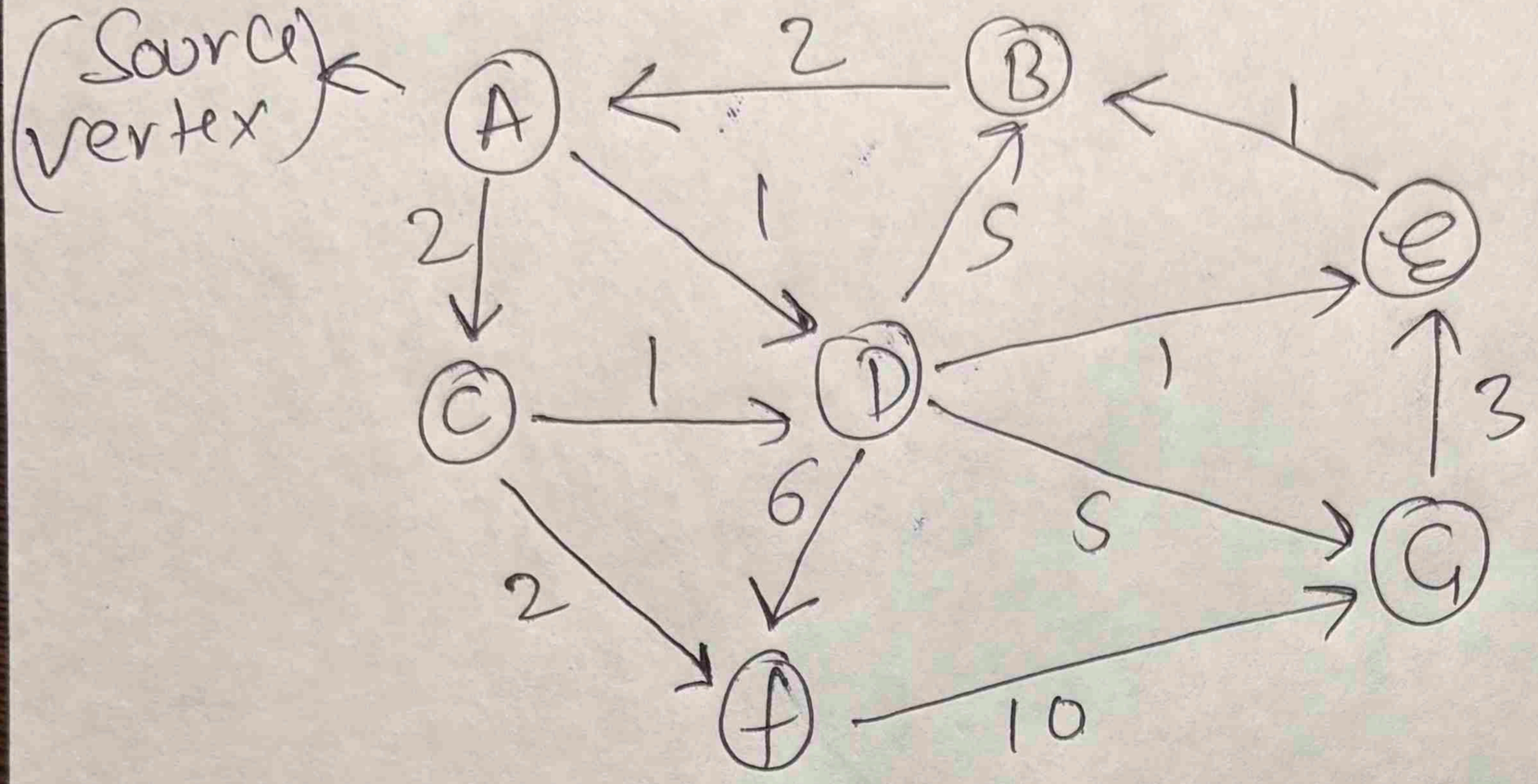
{C-D, B-D, D-E, A-B, D-F, F-G, G-H, H-J, F-I}

Total weight :- {4+3+4+5+5+6+11+13+18 = 66}




Ans

Que-16 Using Dijkstra's Algorithm, find the shortest distance from the source node A to all nodes the following weighted directed graph :-




Sol^n :- Vertices :- {A, B, C, D, E, F, G}

Source :- {A}

Step-1 ⇒ Initialize :-

| Vertex | Distance from A | Previous | Visited |
|---|---|---|---|
| A | 0 | — | NO |
| B | ∞ | — | NO |
| C | ∞ | — | NO |
| D | ∞ | — | NO |
| E | ∞ | — | NO |
| F | ∞ | — | NO |
| G | ∞ | — | NO |

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Step-2 Start from A :-

Edges from A :-

- A → B = 2 → B = 2
- A → C = 2 → C = 2
- A → D = 1 → D = 1

Smallest unvisited Distance (D=1)

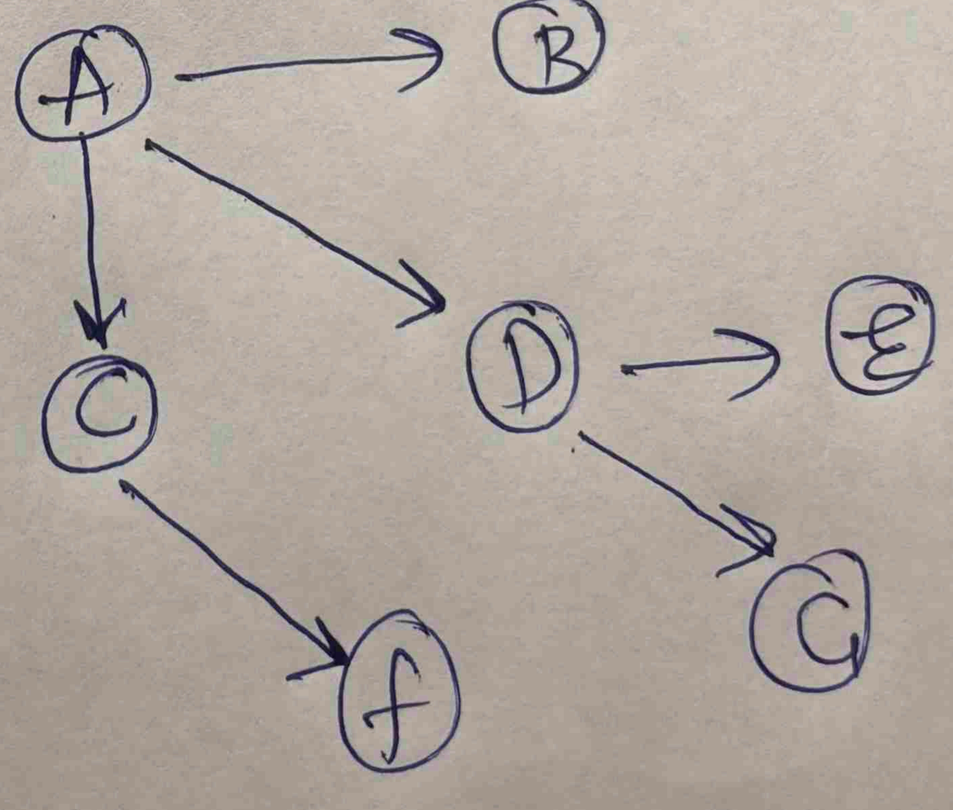
Step-3 Repeat the same process from (B), from (C), from (E), from (F), from (G)

Final Sorted Distance :-

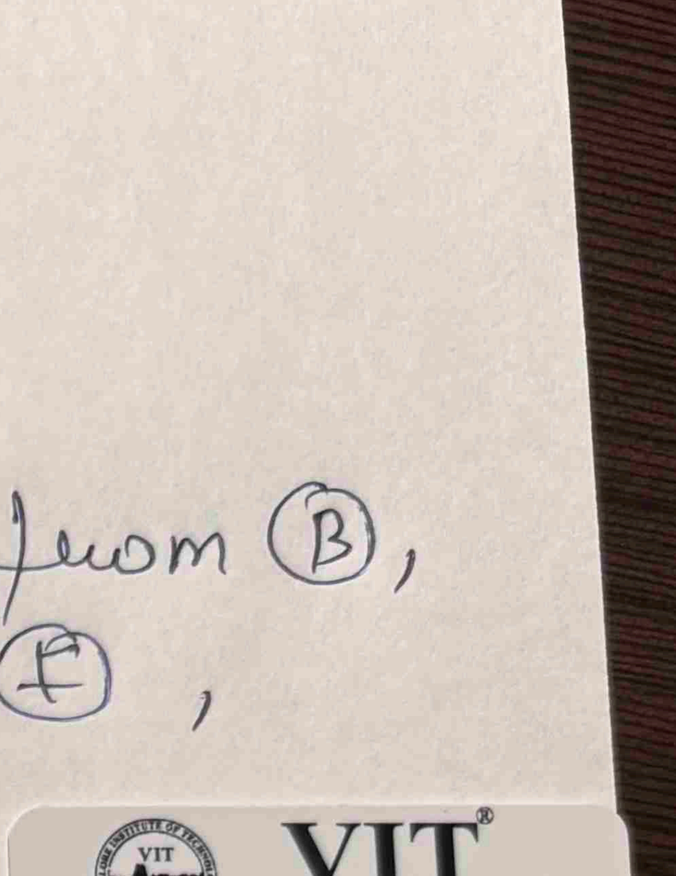
{ A=0, B=2, C=2 | P=1, E=2, F=4, G=6 }

Shortest paths :-

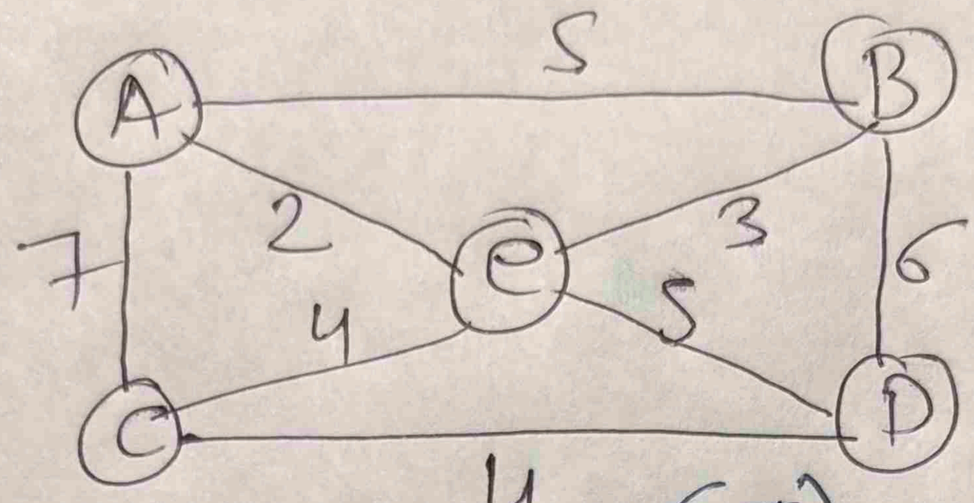
A - B
A - C
A - D
A - D - E
A - C - f
A - D - G




{ final graph } Ans 2



Que-18 Define cut sets and fundamental circuits of a graph. for the following graph, find the cut sets and fundamental circuits. (17)




(T)

Solve :- Chosen Spanning Tree :- {(a,

{Chords (C)} :- {(a,b), (a,c), (b,d

The fundamentals Circuits :-

(formed by Adding one chord to t

- for chord (a,b) : {(a,b), (a,e), (e,b)}
- for chord (a,c) : {(a,c), (a,e), (e,c)}
- for chord (b,d) : {(b,d), (b,e), (e,c), (e,d)}
- for chord (d,e) : {(d,e), (d,c), (c,e)}}

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## Fundamental Cut Sets :-

(formed by Removing one Branch from the tree)

(1) for Branch (a,e) : {(a,e), (a,b), (a,c)}

(2) for Branch (b,e) : {(b,e), (a,b), (b,d)}

(3) for Branch (c,e) : {(a,e), (a,c), (b,d), (d,e)}

(4) for Branch (c,d) : {(c,d), (b,d), (d,e)}

Therefore it follows a cut sets fundamental circuit Any